

Dehydration of 2-Methylcyclohexanol: A Written Lab Report

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Abstract

In this experiment, my objective was to dehydrate 2-Methylcyclohexanol to the Zaitsev major product 1-Methylcyclohexene via the acid catalyst Montmorillonite K10 clay, and I expected to retrieve appropriate H-NMR graphical data, a mid-ranged percent yield, and high purity. In general, the reagent was refluxed near its boiling point to produce the alkene product, which was fractionally distilled through a macroscale setup to retrieve a final sample for H-NMR testing. It was determined that 40.1% percent was returned with mixed H-NMR findings. Finally, it was concluded that the final product was slightly a mixture between Zaitsev major product and Hoffman's minor product, but the percentage yield and mixed results were in reason and somewhat expected.

Introduction

This report encloses the dehydration of 2-Methylcyclohexanol and the importance of green chemistry when conducting laboratory experiments. During the experiment, 2-Methylcyclohexanol was refluxed in a proton-donating, acid catalyst to push the E1 beta-elimination reaction toward the dehydrated products – three possible could be formed (*figure 1*). Through this reaction, the alcohol was transformed into an alkene, by which H₂O was expelled as a biproduct of the reaction; hence, why it's called dehydration. The dehydration reaction generally follows the mechanism of protonating the hydroxy group to increase its leaving group favorability in which the conjugate base deprotonates the beta-carbon's hydrogen. After the beta-carbon obtains electrons from losing its hydrogen, it will undergo an elimination where the additional electrons form a double bond that ultimately expel the H₂O leaving group (*figure 2*). Also, it's important to mention that the dehydration of 2-Methylcyclohexanol has regio-chemistry, meaning it can favor the Zaitsev product (most substituted double bond) or Hoffman product (least substituted double bond), as shown in the mechanism.

Furthermore, to prevent the generation of hazardous substances, green chemistry technique is utilized to prevent toxic Bronsted acid catalysts, like H₂SO₄, from causing issues. In this experiment, I utilized Montmorillonite K10 clay, which acts as a Lewis acid catalyst to perform regenerative protonation/deprotonation of hydrogens instead of such common Bronsted acids.

The purpose of this experiment was to convert 2-Methylcyclohexanol to 1-Methylcyclohexene via a dehydration reaction. In short, during the experiment, I utilized reflux techniques to drive the reaction toward the dehydration product. After, the mixture was fractionally distilled with the addition of PEG-400 to allow higher boiling temperature until about 2.0 mL of Zaitsev product was retrieved. From there, calculations and computerized graphs were made.

For this experiment, I expect to retrieve appropriate graphic H-NMR spectra that matches the Zaitsev and Hoffman products. I hypothesize that the percentage yield will be mid-ranged because of the set amount of product gathered opposed to all possible product, but it will be highly pure.

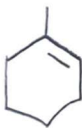
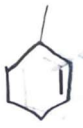
Product 1	Product 2	Product 3
		
IUPAC Name	IUPAC Name	IUPAC Name
<u>1-methylcyclohexene</u>	<u>3-methylcyclohexene</u>	<u>Methylenecyclohexane</u>
Zaitsev or Hofmann	Zaitsev or Hofmann	Zaitsev or Hofmann
<u>Zaitsev</u>	<u>Hofmann</u>	<u>Hofmann</u>

Figure 1: Three Products of 2-Methylcyclohexanol Dehydration

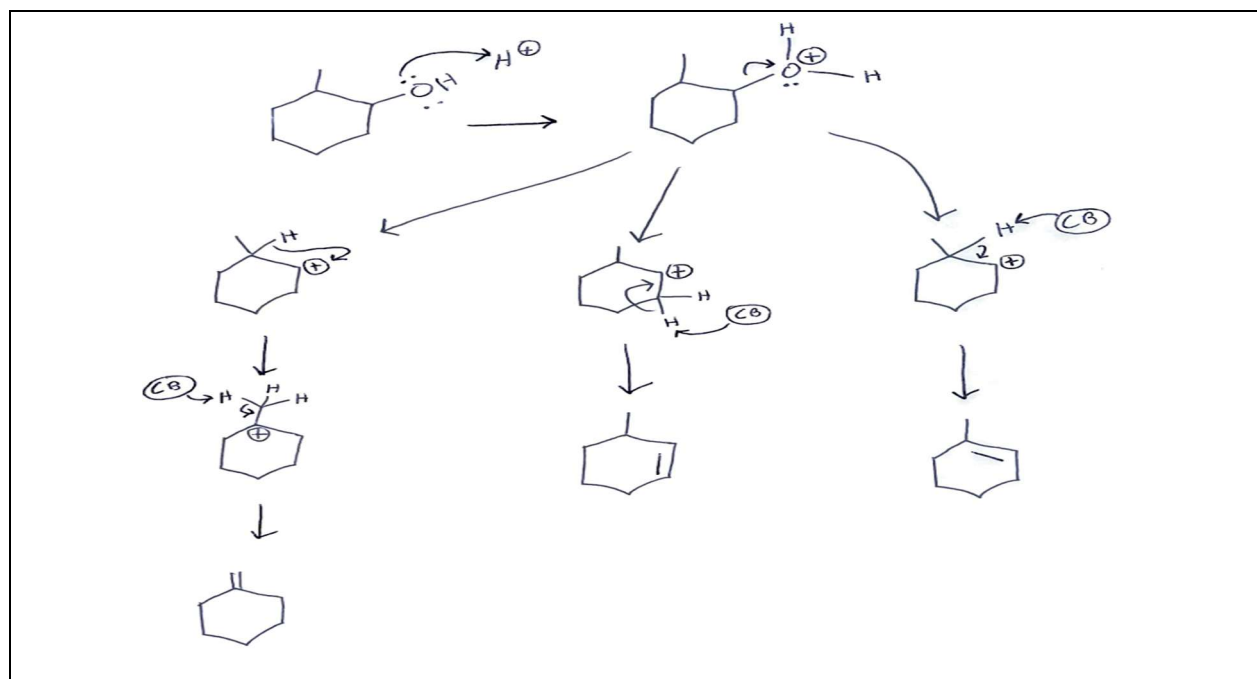


Figure 2: Mechanism of Dehydration of 2-Methylcyclohexanol

Procedure

The procedures of this experiment are as follows:

Materials

A list of utilized chemicals:

- Montmorillonite K10 Clay
- 2-Methylcyclohexanol
- PEG-400
- Anhydrous Sodium Sulfate

A list of utilized equipment:

- Macroscale distillation set-up.

Equations

- Percent Yield = $\left(\frac{\text{Actual Yield}}{\text{Theoretical Yield}} \right) \times 100$

Methods

The general methodology for this experiment is as follows. First, I added 5.0 mL of 2-Methylcyclohexanol and 0.25 grams of Montmorillonite K10 clay to a round bottom flask with a magnetic stir bar. From there, I set up a reflux apparatus with the water hose input from the bottom of the condenser and drain hose output from the top. Then, I refluxed for 90 minutes at around 170°C. After refluxing, I added 3.0 mL of PEG-400 to the mixture that will allow it to be heated to a high temperature to boil the alkenes with solvent remaining, strictly distilling product. Then, I setup a macroscale distillation apparatus (with input and output hose tubes in similar order), and fractionally distilled 2.0 mL of product by one drop per second at roughly 105°C. Finally, I dried the product with sodium sulfate and filtered the Montmorillonite K10 clay out for recycling.

Results

Table 1: Reagent Table

Reagent/Catalyst	Molar Mass (g/mol)	Quantity (mL) - if applicable	Quantity (Grams) - if applicable	Density (g/mL)	Moles
2-Methylcyclohexanol	114.19	5.10	4.74	0.93	0.0415
K10 Clay	360.31	N/A	0.2520	N/A	0.0006994
PEG-400	400	3.00	N/A	1.13	0.00848

Calculation 1: Example Math for Calculating Grams of 2-Methylcyclohexanol

$$\frac{5.10 \text{ mL}}{1} \times \frac{0.93 \text{ g}}{1.0 \text{ mL}} = 4.74 \text{ g}$$

Calculation 2: Example Math for Calculating Moles of 2-Methylcyclohexanol

$$\frac{5.10 \text{ mL}}{1} \times \frac{0.93 \text{ g}}{1.0 \text{ mL}} \times \frac{1.0 \text{ mol}}{114.19 \text{ g}} = 0.0415 \text{ mol}$$

Table 2: Product Yield

Product Yield	Mass (grams)	Moles (mol)
Actual	1.60	0.0166
Theoretical	3.99	0.0415

Calculation 3: Example Math for Calculating Grams and Moles of Actual Yield

$$\frac{2.00 \text{ mL}}{1} \times \frac{0.80 \text{ g Product}}{1.0 \text{ mL Product}} = 1.60 \text{ g}$$

$$\frac{1.60 \text{ g}}{1} \times \frac{1.0 \text{ mol Product}}{96.17 \text{ g Product}} = 0.0166 \text{ mol}$$

Calculation 4: Example Math for Calculating Product Theoretical Yield

$$\frac{0.0415 \text{ mol } 2 - \text{Methylcyclohexanol}}{1} \times \frac{96.17 \text{ g}}{1.0 \text{ mol}} = 3.99 \text{ g}$$

Table 3: Product Percent Yield

Product	Percent Yield (%)
1-Methylcyclohexene	40.1

Calculation 5: Example Math for Calculating Product Percent Yield

$$\left(\frac{1.60 \text{ g Actual}}{3.99 \text{ g Theoretical}} \right) \times 100 = 40.1\%$$

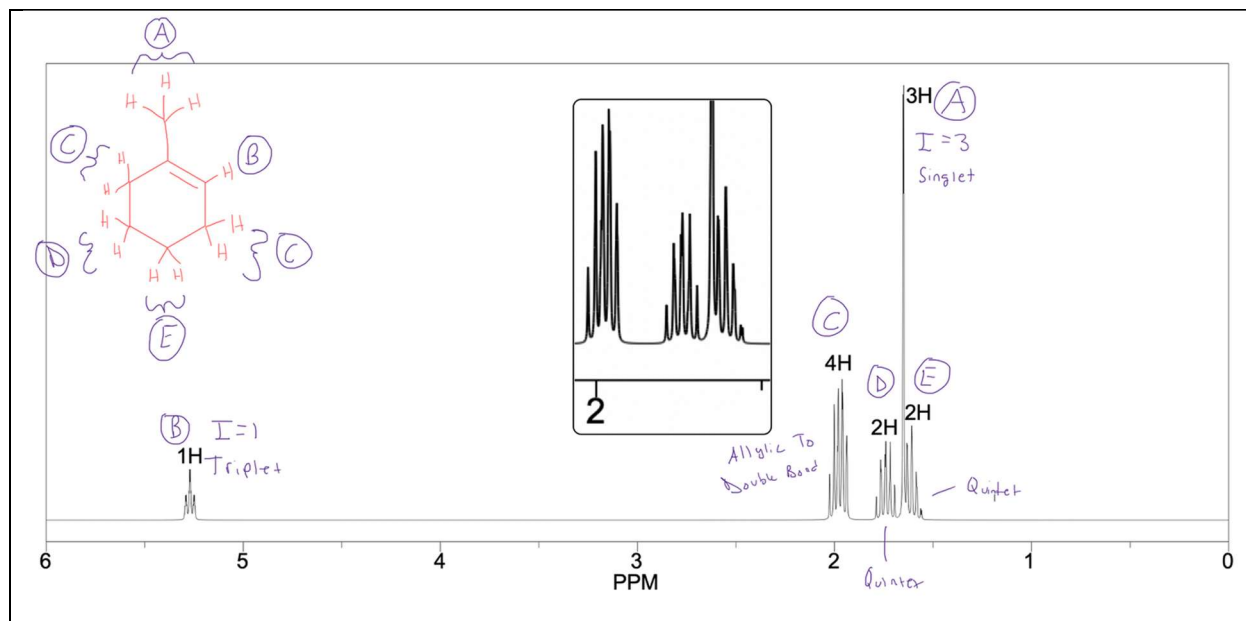


Figure 3: Annotated ^1H -NMR for Major Product (1-Methylcyclohexene) – Non-retrieved

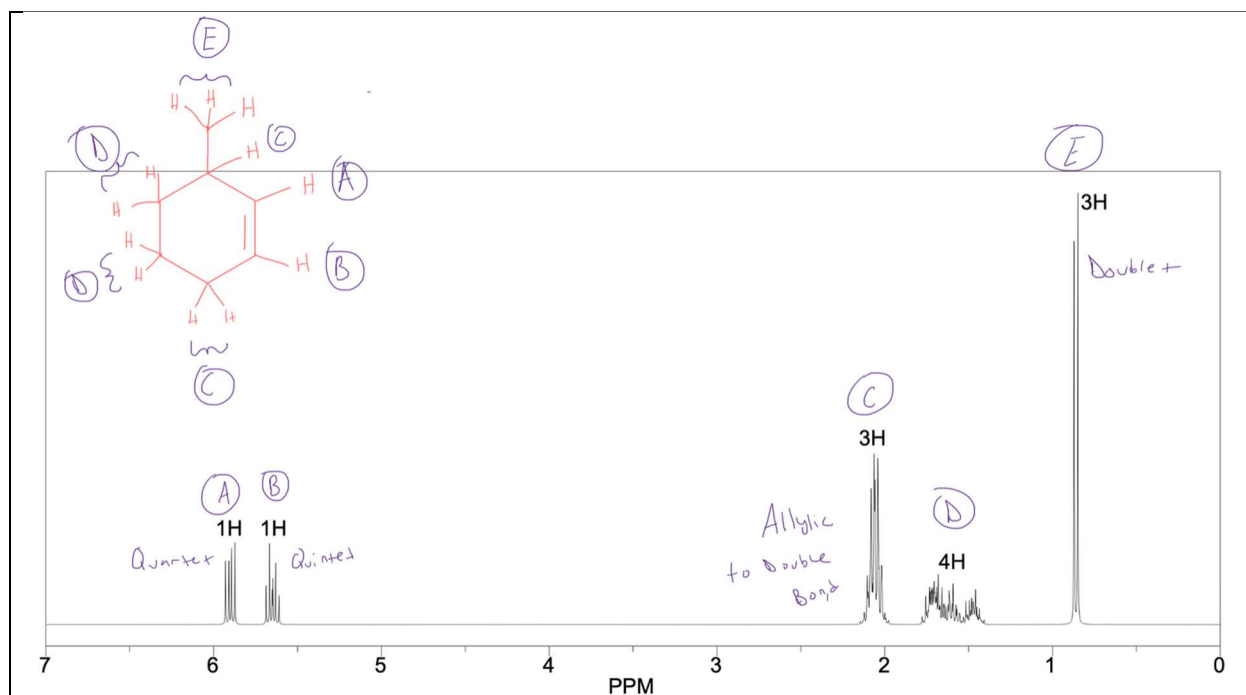


Figure 4: Annotated ^1H -NMR for Minor Product (3-Methylcyclohexene) – Non-retrieved

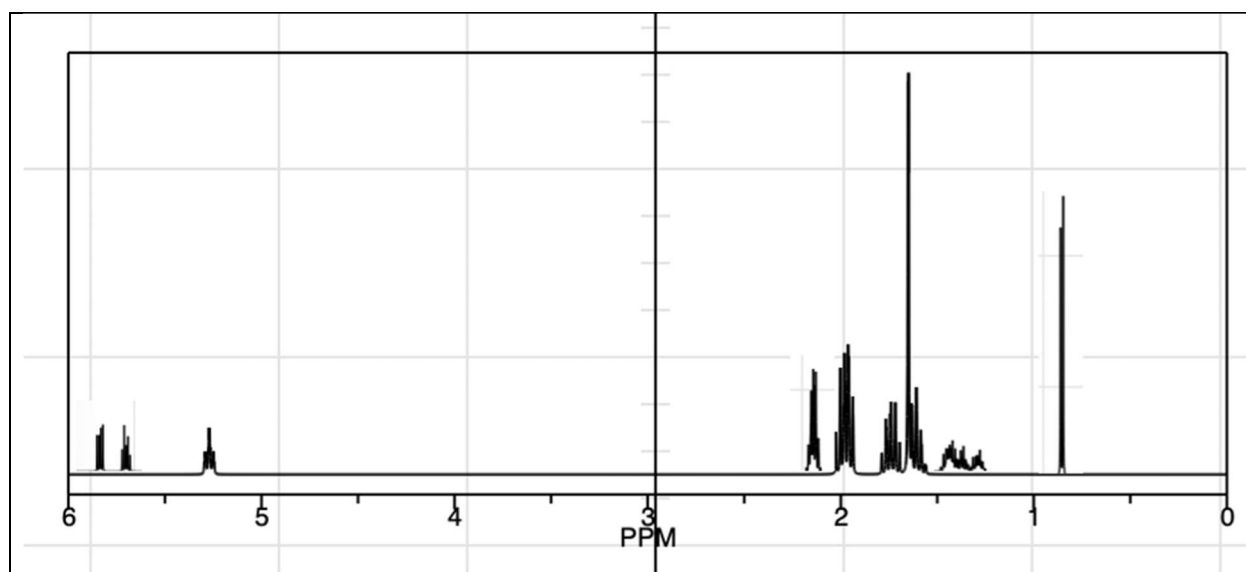


Figure 5: ^1H -NMR for Product retrieved

Discussion

This discussion demonstrates the significance of the results and important findings. My goal of this experiment was to dehydrate 2-Methylcyclohexanol utilizing reflux and fractional distillation with K10 Clay to form the Zaitsev major product (1-Methylcyclohexene). After the procedure, 2.00 mL of alkene product was collected from the macroscale distillation which was calculated to 1.60 g and 0.0166 mol (*Table 2*). With this, a 40.1% percent yield was retrieved with respect to 3.99 g of theoretical product (*Table 3*). The percent yield was moderate but agreed with my expected results that it would mid-ranged due to a limited 2.0 mL distillation, which is lower than the possible, maximum retrieval.

Furthermore, the product H-NMR in *figure 5* showed a variety of peaks at several different shift values that are consistent with both the Zaitsev major product and Hoffman minor product, suggesting that my sample was a mixture between both possible structures and not entirely pure 1-Methylcyclohexene. To confirm this, *figure 5* was compared to sample annotated H-NMR graphs of both the Zaitsev and Hoffman products (*figures 3 and 4*), which clearly indicate that *figure 5* is a mixture because peaks from both graphs are apparent in the H-NMR. However, the graphical pattern for my product is more consistent with the Zaitsev major product (1-Methylcyclohexene) than the Hoffman minor product, as there are only minor contributions from the Hoffman product, but the whole Zaitsev spectrum was shown. In regard to my expectations, I expected that a highly pure substance would be a result, but this was not the entire case with some impurities sweeping by. Nevertheless, a regiochemical mixture is expected when performing dehydration reactions, so it's not a surprise that some managed to become product, and it is consistent with my expectation that the H-NMR would demonstrate both products.

In regard to possible errors, there could have been a few reasons to account for the unexpected results. First, the yield was expected to be mid-range, which is slightly above the actual yield. However, due to a 2.0 mL limited distillation, it's normal for the percent yield to be lower because the actual yield was decreased from what possibly it could have been if the experiment was conducted without restrictions. Additionally, because the boiling points of the regiochemical isomers are close, my fractional distillation could have distilled multiple isomers of the alkene product, which explains the lower purity from the H-NMR perspective (mixed graph). Finally, there could have been minor leaks within the overall setup that could have skewed with some mixture compositions.

Note: All calculations are relative to the table above the calculation figure and explain how each operation was performed.

Conclusion

In summary, this experiment has succeeded in demonstrating the dehydration of 2-Methylcyclohexanol to 1-Methylcyclohexene as the major product. Additionally, this experiment supported my hypothesis that the H-NMR spectrum would demonstrate both Zaitsev and Hoffman products with a generally mid-ranged percent yield; however, my expectations of a highly pure product were contradicted by my sample, which contained some significant impurities of the minor products. Regarding possible errors in the experiment, the results were a mix of human and other errors, such as a limited number of extracted products below a possible, experimental actual yield and close boiling points of all three products, which may have distilled more of the minor product than intended. Finally, for the future, I would suggest that all of the major product should be extracted from the distillation to get the most accurate percent yield, which would verify the strength of the reflux mechanism.